

Multi-sensor dataset of a tendon-driven continuum robot in dynamic motion

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Abstract

This data descriptor presents a multi-sensor dataset of a tendon-driven continuum robot undergoing quasi-static and dynamic motions, including scenarios with contact in the workspace. The dataset provides synchronized measurements of robot kinematics, actuation states, and interaction forces across multiple sensing modalities, covering planar, circular, and Lissajous trajectories executed at different speeds. Both raw sensor recordings and processed data (filtered, time-aligned, and uniformly resampled) are included. To ensure data quality and internal consistency, multiple cross-validation procedures are performed across sensing modalities, including comparisons between independent kinematic measurements and consistency checks between force and motion data. To our knowledge, this is one of the first publicly available datasets that combines these modalities within a unified experimental framework for a continuum robot. The dataset supports benchmarking of dynamic models, development of data-driven methods, analysis of contact behavior, and sensor fusion research. All data are provided in structured, machine-readable formats with detailed documentation, facilitating direct reuse and reproducibility.

1 Background & Summary

Continuum robots are manipulators whose body undergoes continuous elastic deformation, actuated by tendons, pneumatics, or other distributed mechanisms [1, 2]. Their dynamics have been studied through various modeling approaches [3, 4, 5, 6], and there is growing interest in applying machine learning methods, in both statics [7] and dynamics [8], as well as studying their contact and interaction with the environment [9] with new studies addressing the contact kinematics with differentiable properties [10]. However, experimental validation in continuum robotics typically relies on ad-hoc prototypes instrumented with lab-specific sensors, and the resulting data is rarely made publicly available. A related effort produced a dataset for concentric tube robots [11], but no publicly available dataset currently combines kinematics and force measurements for a tendon-driven continuum robot undergoing both quasi-static and dynamic motions, including contact with objects in the workspace. This limited availability of open datasets prevents reproducibility of results and the comparison of competing modeling, control, and learning approaches on common ground.

This data descriptor presents a multi-sensor dataset of a single-segment tendon-driven continuum robot undergoing quasi-static and dynamic motions. The dataset provides synchronized measurements of robot kinematics, actuation states, and interaction forces acquired from multiple sensing modalities. Kinematic information includes motion capture of disk poses, distributed shape sensing from fiber Bragg grating sensors (FBGS), and actuator encoder readings, while force-related measurements include tendon tensions, base wrench, and contact forces during interaction with external objects. Collectively, these measurements span a broad range of state variables commonly considered in the continuum robotics literature.

A key feature of the dataset is the integration of these heterogeneous sensing modalities within a unified experimental framework, enabling direct cross-modal analysis. To ensure data quality and internal consistency, cross-validation procedures are performed across sensing modalities, including comparisons between independent kinematic measurements and consistency checks between force and motion data.

The dataset is organized into three subsets corresponding to distinct experimental conditions: (i) quasi-static motions, (ii) dynamic motions, and (iii) dynamic motions with contact. The dynamic subset

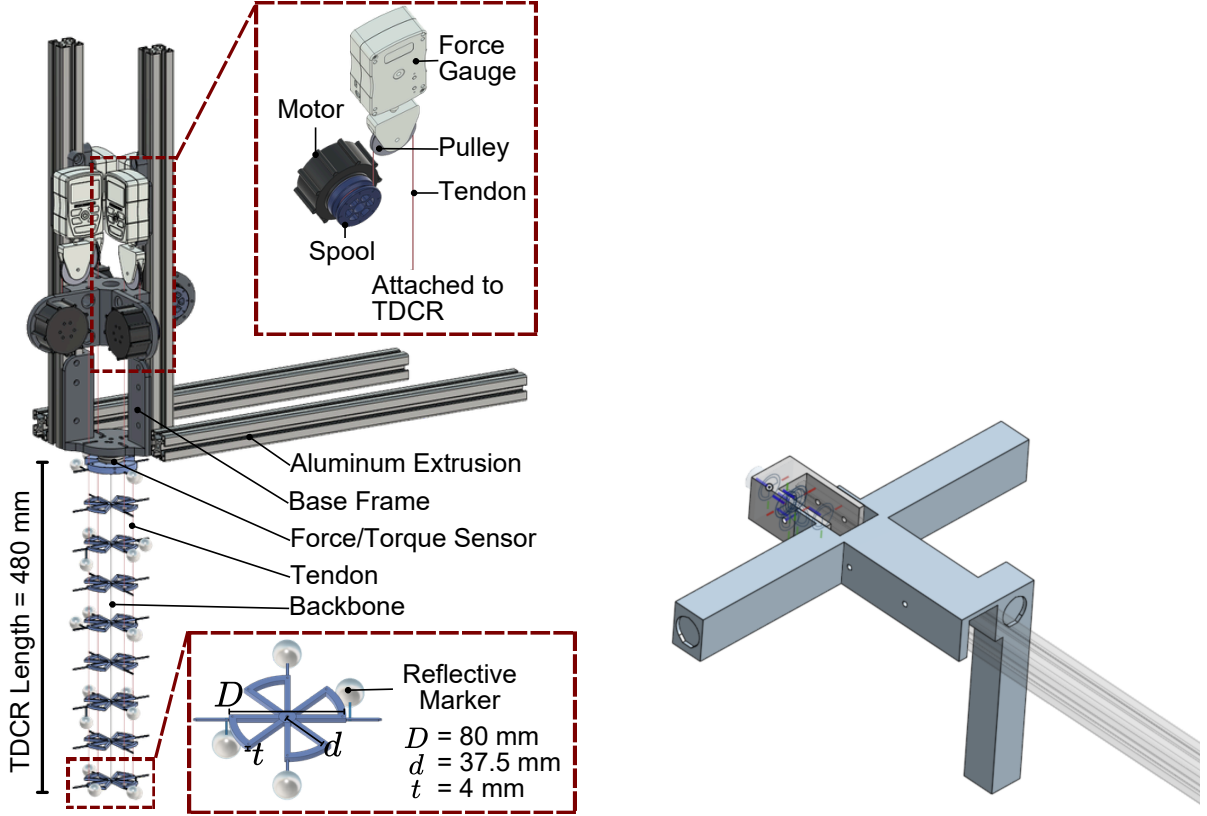


Figure 1: Design of the Tendon-Driven Continuum Robot (left) and the contact wand (right).

includes a range of representative trajectories, including planar sweeps, star patterns, circular paths, and Lissajous curves, executed at two actuation speeds. For each experiment, both raw sensor recordings and processed data (filtered, time-aligned, and uniformly resampled) are provided, together with detailed documentation of file structure and variables.

This dataset is designed to support a broad range of research tasks, including benchmarking of continuum robot dynamics models, development and evaluation of data-driven approaches, analysis of contact interactions, sensor fusion, and parameter identification. **By combining multi-modal sensing, dynamic motion, and contact interactions in a single dataset, it provides a comprehensive resource for studying continuum robot behavior under realistic operating conditions.** Furthermore, the use of commercially available components and fully documented experimental procedures facilitates reproducibility and enables other researchers to replicate or extend the dataset on similar platforms.

2 Methods

In this section, we define the methodology used to create the presented dataset, starting from the robot design and components to the specifics of the sensors. Finally, we describe how the data is collected ~~giving a general idea~~ of the data collection pipeline.

2.1 Design

In this section we describe both the design of the continuum robot and the contact wand that is used to interact with the robot.

2.1.1 Continuum robot

The experimental platform is designed to enable high-fidelity and reproducible measurement of both kinematic and force quantities by minimizing sources of systematic error such as friction, structural

compliance, and geometric uncertainty. The system consists of a bench-top experimental platform and a tendon-driven continuum robot (Fig. 1).

The platform provides a rigid and geometrically stable structure for actuation, sensing, and robot mounting. It includes a 3D-printed base (PETG), four aluminum extrusion frames (40 mm T-slot profiles), and custom mounting adapters for the motors and force gauges. The vertical and horizontal frames are rigidly joined using T-slot connectors, forming a stable structural frame mounted on a leveled table to ensure a consistent gravity reference across experiments. Actuation components are arranged symmetrically about the robot centerline to reduce unintended loading and improve repeatability of the recorded measurements.

The robot consists of a slender NiTi tubular backbone that accommodates internal sensing. Material properties are reported in Table 1. Nine equally spaced spacer disks (6 cm spacing) are mounted along the backbone and provide attachment points for motion capture markers and tendon routing. Each disk is instrumented with a marker configuration enabling full six-degree-of-freedom pose reconstruction within the motion capture system, with unique marker layouts used to facilitate robust tracking. Four tendons are attached at the distal disk and routed proximally through the disks with an approximately constant radial offset of 37.5 mm, enabling controlled planar and spatial deformations.

At the base, each actuator is paired with an inline force gauge to directly measure tendon tension. Tendons are redirected using pulley adapters mounted on the sensing interface of the force gauges, and no intermediate pulleys are used between the robot and the sensors, ensuring that measured forces correspond directly to applied tendon loads. The four actuator-sensor pairs are labeled 1–4, corresponding to tendons aligned with the x , y , $-x$, and $-y$ directions. A six-axis force-torque sensor is mounted between the base platform and the robot base disk to measure boundary forces and moments. The mounting interface is reinforced to minimize compliance and preserve measurement accuracy.

Overall, the platform is designed to provide consistent, low-noise, and physically interpretable measurements across all sensing modalities, supporting reliable data collection for benchmarking, model validation, and data-driven analysis.

2.1.2 Contact wand

The contact wand provides a lightweight slender object for interacting with the robot without occluding the disk markers (Fig. 1). It consists of an aluminum extrusion frame (20 mm T-slot profile) equipped with a motion capture rigid body (for 6-DoF pose tracking) and a force/torque sensor for measuring the interaction force. A custom aluminum adapter aligns the sensor axes with those of the motion capture frame.

The pose of the Resense sensor frame relative to the wand motion capture frame is described by a fixed rigid-body transformation $g_{\text{fix}} \in SE(3)$, measured once during assembly and used in post-processing to transport the contact wrench to any desired frame. The rotational part is composed of a rotation of 90° about the x -axis followed by a rotation of 30° about the z -axis, accounting for the physical misalignment between the sensor connector orientation and the motion capture frame axes. The translational offset is $\mathbf{r}_{\text{fix}} = (0, -0.1137, 0)^T \text{m}$, corresponding to the distance along the wand body between the motion capture frame origin and the Resense sensor centre. The contact wrench measured at the sensor frame can be transported to any frame of interest via $\mathcal{F} = \pm \text{Ad}_g \mathcal{F}_{\text{sensor}}$, where $g = {}^{ref}g_{wand}g_{fix}$ is the rigid-body transformation from the desired reference frame to the sensor frame, with ${}^{ref}g_{wand}$ the pose of the wand with respect to the reference frame. As an example, transporting the wrench to the robot base frame yields the contact contribution to the base wrench, which can be compared directly with the ATI-FT measurements.

Table 1: Properties of the NiTi tubular backbone.

Length ℓ	480 mm
Outer diameter d_e	2.019 mm
Inner diameter d_i	1.487 mm

Table 2: Tendon Actuation.

N tendons	4 (2 pairs of antagonist cables)	
Pulleys diameter r_s	20 mm	
Distance to backbone d_c	37.5 mm	
Material	Polyethylene (braided)	
Brand	Piscifun	
Specifications	12 lb	0.12 mm

2.2 Sensors

The robot is instrumented with sensors capturing both kinematic and force quantities: position and shape (motion capture, FBGS, motor encoders) as well as tendon tensions, base wrench, and contact forces. This section describes each sensor in terms of sampling rate, measurement range, and key specifications from the manufacturer datasheet.

2.2.1 Mark 10

Cable tension was measured using Mark-10 M3-5 digital force gauges (Mark-10 Corporation, Copiague, NY, USA; <https://mrmmetrology.com/product/m3-5-mark-10-digital-force-gauge/>), with a full-scale capacity of 5 lbf (≈ 22.2 N), a resolution of 0.005 lbf (≈ 0.02 N), and a manufacturer-specified accuracy of $\pm 0.3\%$ of full scale. All units were calibrated by an ISO 17025 accredited laboratory with an expanded uncertainty of ± 0.007 lbf ($k=2$, 95% coverage). Each gauge was interfaced to the acquisition computer via a serial connection at 115 200 baud and polled continuously using a request-response protocol. The effective sampling rate, limited by the serial round-trip latency, was approximately 335 Hz per gauge. The output was recorded directly in newtons. A dedicated Python script (one instance per gauge) opens the serial port via `pyserial`, sends an ASCII query command, parses the returned numeric string, and timestamps each reading with the system wall clock (`time.time()`). All samples are buffered in a pre-allocated NumPy array and written to a CSV file at the end of the recording.

2.2.2 FBGS

Distributed strain along the robot backbone is measured with an FBGS four-core multicore fiber (MCF, ID: F13421-0007-003; FBGS International, Belgium) featuring 26 draw-tower gratings per core at 20 mm spacing, for an active sensing length of 500 mm. Three cores are used for shape reconstruction (core 2 omitted), enabling independent estimation of bending curvature, bending direction, and temperature at each grating.

The fiber is interrogated by an FBG-Scan 904 spectrometer (1510 nm to 1590 nm, 4 channels, wavelength precision 0.3 pm 1σ). ILLumiSense Shape CORE v1.1 performs peak detection and outputs the reconstructed 3-D shape at 1 mm arc-length resolution and 100 Hz. A C++ client receives the TCP stream, timestamps each sample with `std::chrono::high_resolution_clock`, and writes the buffered data to CSV after the recording.

Prior to use, the distal grating was located to ~ 1 mm via a hot-wire contact procedure and aligned with the tip disk; the fiber is fixed at the base and tip to avoid twisting inside the backbone.

2.2.3 ATI-FT

External force and torque measurements at the robot base were acquired using an ATI Mini40 six-axis force/torque sensor (calibration SI-40-2; ATI Industrial Automation, Apex, NC, USA; https://www.ati-ia.com/products/ft/ft_models.aspx?id=mini40). With the SI-40-2 calibration, the sensing ranges are ± 40 N in F_x and F_y , ± 120 N in F_z , and ± 2 Nm in T_x , T_y , and T_z . Single-axis overload ratings are ± 810 N (F_{xy}) and ± 2400 N (F_z), and the resonant frequencies are 3200 Hz (F_x , F_y , T_z) and 4900 Hz (F_z , T_x , T_y). The transducer was connected in the DAQ F/T configuration, with the six analog voltage channels (± 10 V range) digitized by a National Instruments data acquisition card using the `nidaqmx` Python driver. The hardware sampling clock was set to 1000 Hz; every five consecutive samples per channel were averaged to yield an effective wrench output rate of approximately 200 Hz. A

6×6 calibration matrix provided by ATI was applied in software to convert the bias-subtracted voltage readings into calibrated force and torque values ($F_x, F_y, F_z, T_x, T_y, T_z$).

2.2.4 Actuators

The tendons are actuated by four brushless DC motors (Cybergear, Xiaomi, China) with custom 3D-printed tendon spools of winding radius 20 mm, i.e., the radius at which the tendon cable is coiled around the spool. Each motor is a quasi-direct-drive (QDD) module integrating a three-phase, 28-pole frameless torque motor with a 7.75:1 planetary gearbox, a 14-bit single-turn absolute encoder, and onboard FOC electronics, in a compact package of 80.5 mm diameter and 36.5 mm height (317 g). The motors are powered at 24 V DC and provide a continuous output torque of 4 Nm and a peak torque of 12 Nm. Communication with the host PC is handled via a CAN 2.0 bus at 1 Mbits s^{-1} . The four motors are operated as two antagonistic pairs (motors 1&3 and 2&4), each pair controlling one planar degree of freedom of the robot tip. They are operated in a closed-loop position control scheme, where target shaft angles θ^* are commanded directly over CAN, with shaft angle, velocity, and output torque returned as real-time feedback at each control cycle. Angles are computed relative to the home position, which is initialized as the reading angle at the motor startup. The CAN bus is accessed from the host PC through a USB-CAN adapter (Canable / candle interface) using the `python-can` library. A custom Python script, built on the open-source Cybergear driver [12], instantiates four `CANMotorController` objects, sets each motor to position mode, and commands target angles in a sequential loop. At each control cycle the script reads back the shaft angle, velocity, and torque from every motor and timestamps the sample with the system wall clock (`time.time()`). The feedback data is buffered in memory and saved to a single CSV file at the end of the recording.

2.2.5 Motion Capture System

The shape of the robot backbone and the pose of objects in contact with it are measured using an optical motion capture system (OptiTrack, NaturalPoint, USA) composed of five PrimeX13 and two PrimeX13W cameras. The PrimeX13 cameras feature a 1.3 MP sensor ($1280 \times 1024 \text{ px}$), a native frame rate of 240 Hz, a field of view of $56^\circ \times 46^\circ$, and a 3D positional accuracy of $\pm 0.15 \text{ mm}$. The two PrimeX13W cameras share the same sensor and frame rate but provide a wider field of view of $82^\circ \times 70^\circ$, suited to ensure full coverage of the compact workspace, at a slightly reduced accuracy of $\pm 0.30 \text{ mm}$. All cameras use 850 nm infrared illumination and communicate via GigE/PoE. The seven cameras were arranged around the robot at a working distance of approximately 2 m, providing full coverage of the $1 \times 1 \times 1 \text{ m}$ workspace. Five rigid bodies composed of passive retro-reflective markers (6 mm diameter) are attached to equally-spaced disks along the robot backbone at arc-length positions $s \in \{0, 12, 24, 36, 48\} \text{ cm}$, providing discrete samples of the backbone pose. An additional rigid body is mounted on the contact object to track its 6-DoF pose within the workspace. The system runs on a dedicated host PC managed by Motive 3.4, which processes and solves 3D pose at 120 Hz. Data are streamed in real time to the dataset acquisition PC via a point-to-point Ethernet link using the NatNet 4.4 SDK in unicast UDP mode. A custom Python client receives each frame and logs the 6-DoF pose — position (x, y, z) and orientation (q_x, q_y, q_z, q_w) — of each rigid body, together with a per-body tracking validity flag. Timestamps are assigned on the acquisition PC using the system wall clock (`time.time()`), rather than Motive’s internal camera exposure timestamps, which could induce a small and variable network transmission jitter in the logged timing.

2.2.6 Resense FT

Contact forces between the robot and objects in the workspace were measured using a Resense HEX 12 six-axis force/torque sensor (ordering code HEX125S-020S-1R1-1; Resense GmbH, Klingenberg, Germany; <https://www.resense.io>). The nominal measurement ranges are $\pm 25 \text{ N}$ in F_x, F_y , and F_z , and $\pm 125 \text{ mNm}$ in T_x, T_y , and T_z , with a manufacturer-specified accuracy of 1% of full scale, a crosstalk of 3%, and an overload capacity of 300%. The sensor was paired with the Resense miniature evaluation electronics board (EVAL 100S-06-1, $54 \times 25 \times 12.6 \text{ mm}$), which digitises the strain gauge signals with a 24-bit ADC, applies the manufacturer-provided calibration matrix onboard, and outputs calibrated forces (in newtons) and torques (in millinewton-metres) as seven 32-bit floating-point values per sample ($F_x, F_y, F_z, T_x, T_y, T_z$, and temperature). The electronics was connected to the acquisition computer via its USB interface (12 Mbit/s) appearing as a virtual COM port, and configured at a sample rate of 100 Hz. A custom Python script reads the 28-byte binary packets from the serial stream and logs each

sample with a wall-clock timestamp. This sensor was used exclusively in subset 3 (dynamic motions with contact).

Table 3: Summary of the sensor suite. Effective sampling rates account for communication overhead and software processing; native hardware rates may be higher. Selectable parameters show the value used in this work where known; the full set of options is given in the corresponding sensor subsection.

Sensor	Model / Manufacturer	Qty	Measured quantity	Range	Accuracy / Resolution	Interface	Eff. rate	Subsets
Cable tension	Mark-10 M3-5 (Mark-10 Corp., USA)	$\times 4$	Tendon force (1-axis)	0 N to 22.2 N (5 lbf full scale)	$\pm 0.3\%$ FS; res. 0.02 N; cal. unc. ± 0.007 lbf ($k=2$)	Serial RS-232 115 200 baud	~ 335 Hz	1–3
Shape sensing	FBGS FBG-Scan 904 + 4-core MCF fiber (FBGS Int., Belgium)	1 fiber, 26 DTGs/core	Distributed curvature, bending angle, 3-D shape	50 mm active; 1510 nm to 1590 nm	WL prec. 0.3 pm (1σ); lin. 3 pm; abs. acc. 10 pm; shape res. 1 mm arc	USB; ILLumiSense Shape CORE v1.1	100 Hz	1–3
Base F/T	ATI Mini40 SI-40-2 (ATI Ind. Autom., USA) + NI USB-6210 DAQ	1	6-axis wrench ($F_x, F_y, F_z, T_x, T_y, T_z$) at robot base	$F_{x,y}$: ± 40 N; F_z : ± 120 N; $T_{x,y,z}$: ± 2 N m	16-bit DAQ (250 kS/s); overload $F_{xy} \pm 810$ N, $F_z \pm 2400$ N	Analog ± 10 V via <code>nidaqmx</code> ; HW clk 1000 Hz, 5-samp. avg.	~ 200 Hz	1–3
Motion capture	OptiTrack PrimeX 13 ($\times 5$) + PrimeX 13W ($\times 2$) (NaturalPoint, USA)	7 cameras; 6 rigid bodies	6-DoF pose (5 disks + 1 object)	Sub-metre workspace; 1.3 MP; 850 nm IR	± 0.15 mm (PrimeX 13); ± 0.30 mm (13W); markers ≈ 6 mm dia.	GigE/PoE; Motive 3.4; NatNet 4.4 SDK (unicast UDP)	120 Hz	1–3
Actuators	Cybergear QDD (Xiaomi, China)	$\times 4$	Shaft angle (14-bit), velocity, torque	Cont. 4 N m; peak 12 N m; spool $r=20$ mm	14-bit single-turn absolute encoder	CAN 2.0 1 Mbit/s	460 Hz	1–3
Contact F/T	Resense HEX 12 (Resense GmbH, Germany) + EVAL 100S-06-1 electronics	1	6-axis wrench ($F_x, F_y, F_z, T_x, T_y, T_z$) at contact point	$F_{x,y,z}$: ± 25 N $T_{x,y,z}$: ± 125 mN m	1% FS; 3% crosstalk; 24-bit ADC; 300% overload	USB (12 Mbit/s) virtual COM; 28-byte binary packets	100 Hz	3

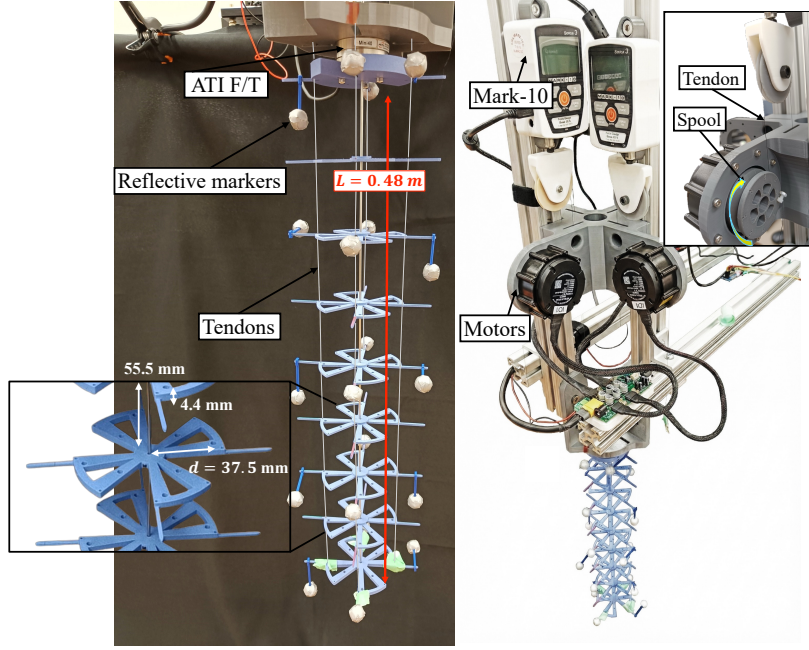


Figure 2: Hardware platform: TDCR with four tendons (two antagonistic pairs), motion capture markers on selected disks, direct-drive motor spools, and inline Mark-10 tension sensing and ATI F/T placed at the base.

2.3 Experimental setup

The robot is mounted on a cantilever platform with its tip pointing downward, *i.e.* in the direction of gravity (see Fig. 2). This layout was chosen as representative of the tendon-driven continuum robots commonly studied in the literature. The same layout is used across all subsets of the dataset. In subsets 1 and 2, the workspace is free of obstacles, while in subset 3 a contact wand is introduced. To ensure consistency across all sensor modalities, every recording session follows a fixed preparation protocol.

First, the FBGS is aligned to the robot motion following the procedure. Using a self-leveling laser (Bosch Professional GLL50-20), the robot is verified straight along both the x and y axes, with cables compensating any residual deformation of the backbone. From this configuration, the ShapeCore software *set_straight* functionality allows us to update the fiber calibration matrix to comply with the registered shape. Then, by pulling on the $+x$ tendon the robot is significantly bent in one direction, with usage of the laser the robot is ensured bent in the xz plane, correcting any out of plane deformation with the remaining cables. From this planar bent configuration, the *set_planar* functionality of ShapeCore ensures that the measurements are free from any residual twisting.

Once the fiber is calibrated, two preliminary recordings are performed. First, with all tendons released, baseline noise is measured on all sensors. Then, with the robot verified straight using the Bosch laser, the straight configuration is recorded to provide an estimate of the residual error in the FBGS and motion capture measurements.

Moreover, before each experiment, the laser is used to ensure that the robot is straight. In this configuration, the active tendons are pre-tensioned to approximately 1 N each, with a maximum difference between the two cables of an antagonistic pair < 0.2 N. This step minimizes the risk of cable slack, a well-known source of modeling error in tendon-driven continuum robots [13].

Finally, for every non-planar trajectory (*i.e.* star, circular and Lissajous curves) the robot executes a calibration routine before the target motion. The routine consists of a planar sweep in the $x-z$ plane (positive x followed by negative x), then a planar sweep in the $y-z$ plane (negative y followed by positive y). Because aligning the FBGS fiber to the robot backbone with sub-degree accuracy is difficult in practice, a residual angular offset between the fiber deformation plane and the robot bending plane is expected. The calibration sweeps characterize this offset, allowing it to be corrected in post-processing before the non-planar data is interpreted.

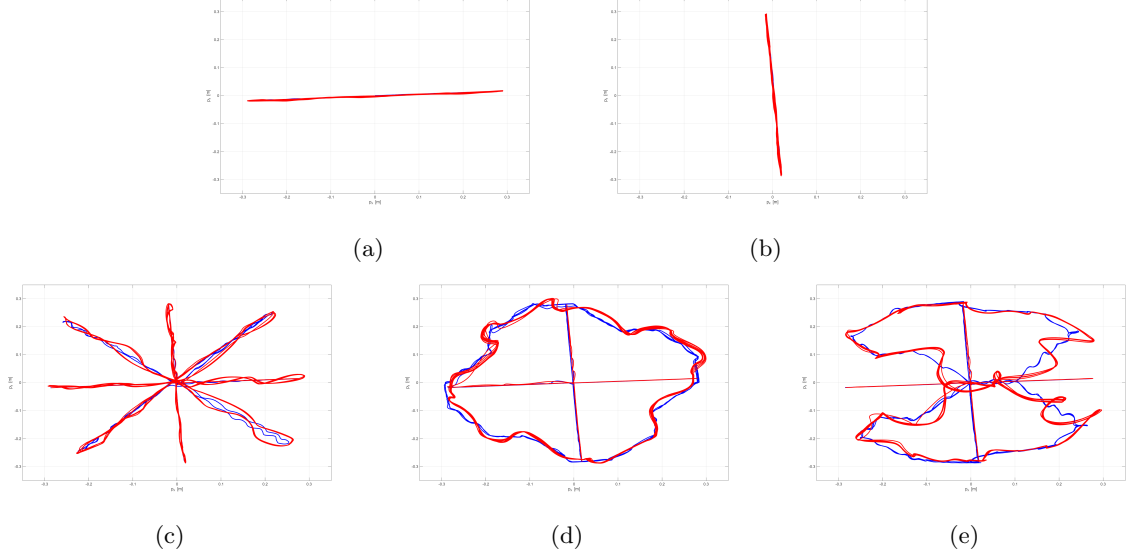


Figure 3: Motion of the robot tip (only xy plane coordinates) for the planar motions on x (a), y (b), the star motion (c), the circular (d) and Lissajous (e) trajectories for the slow (blue) and fast (red) actuators configuration.

2.4 Data Collection

2.4.1 Acquisition pipeline

For each experiment, a Windows batch script orchestrates the acquisition. The script launches one independent process per sensor, each in its own terminal window. Each sensor process polls its instrument at the native sampling rate (see the sensor specifications in Table 3), buffers all readings in memory, and writes them to a CSV file after the recording is complete. The motor process is launched last and waits 2s before sending position commands, ensuring all sensors are already streaming before the actuators begin to move.

Two sensors require dedicated proprietary software that runs independently from the per-sensor Python acquisition scripts. The FBGS fiber is managed by the ILLumiSense ShapeCore software (v1.1, FBGS), which runs as a background process on the acquisition computer, connects to the FBG-Scan 904 interrogator via USB, performs peak detection on the acquired spectra, and reconstructs curvature, bending angle, and three-dimensional shape in real time. ShapeCore streams its output over a local TCP connection. A custom C++ client (**StreamClassTester**) connects to this stream, receives the curvature, angle, and shape arrays at each time step, timestamps each sample (using `std::chrono::high_resolution_clock`), and buffers all data in memory until the recording is complete, at which point it writes a single CSV file. The OptiTrack motion capture system is driven by Motive 3.4 (NaturalPoint, USA) running on a dedicated Windows host PC. Motive processes and solves the 3-D rigid-body poses at 120 Hz and exposes them through the NatNet 4.4 SDK in unicast UDP mode. A dedicated Python script on the acquisition machine receives each frame over a point-to-point Ethernet link and timestamps it with the wall-clock of the acquisition machine (`time.time()`), ensuring a homogeneous time base across all sensor modalities.

2.4.2 Synchronization

All timestamps in the recorded CSV files are therefore referred to the same operating-system clock, without the need for a hardware trigger. The synchronization accuracy is bounded by the OS thread-scheduling jitter, on the order of 1 ms to 2 ms on the acquisition machine.

2.4.3 Post processing

The raw sensor data is post-processed in MATLAB (`process_data.m`).

The samples from the mocap system are converted from position (x, y, z) and quaternion to SE(3) poses. Because these poses are expressed in the global motion-capture frame, they are re-expressed as

transformations relative to the base of the robot. To avoid propagating position and orientation error, we defined a base frame coincident with the position of disk_0 and with axes aligned with the direction of the robot. Then the pose of every disk frame is computed with respect to this common base.

For the FBGS data, the interrogator software reconstructs the fiber shape with the arc-length parameter evolving along its internal x -axis, while the physical robot hangs along negative z ; therefore, a rigid rotation is applied to map the FBGS coordinate frame to the robot body frame. Furthermore, because the fiber cannot be bonded to the backbone with sub-degree angular accuracy, a residual rotational offset exists between the fiber deformation plane and the true robot bending plane. To characterize this offset, a singular-value decomposition is performed on the tip trajectory during the initial planar calibration sweep of each batch: the principal directions of the tip motion identify the bending-plane and the corresponding rotation angle, namely θ_{FBGS} . As the motion of the FBGS has to be aligned with the recorded motion of the mocap, a similar SVD-based procedure is applied on the tip motion as registered by the mocap, yielding θ_{mocap} . The correction applied to the FBGS, as a rotation along the z axis, of the amount $\theta_z = \theta_{FBGS} - \theta_{mocap}$.

All signals are then low-pass filtered with a fourth-order Butterworth filter (cutoff frequency 20 Hz applied in the forward and reverse directions (`filtfilt` in MATLAB) to achieve zero-phase distortion. The filter’s normalized cutoff is computed per signal using each sensor’s native sampling rate, estimated from the median inter-sample interval. The filtered signals are aligned in time by expressing all timestamps relative to the first actuator timestamp; because every sensor starts recording before the motors, these relative timestamps are initially negative. All signals are then resampled onto a uniform 100 Hz grid spanning $t = 0$ to the end of the actuator recording using linear interpolation (`interp1`). This operation simultaneously discards the sensor preamble (the negative-time samples recorded before the motors started) and brings all modalities to a common time base.

2.4.4 File naming and outputs

Each sensor process writes one CSV file per experiment, with naming convention: `data{Sensor}.csv` (e.g. `dataMotor.csv`, `dataMark10+x.csv`, `dataATIFT.csv`, `dataOptiTrack.csv`, `dataFBGS.csv`, and `dataResenseFT.csv` for subset 3). Column headers and units are documented in the Data Records section. The post-processed data is saved in a `processed/` subfolder within each experiment directory, containing one file per quantity (`angles.csv`, `cable_tensions.csv`, `base_wrench.csv`, `mocap_frames.csv`, `fbgs_shapes.csv`), all sampled at 100 Hz with a shared time column. For subset 3 the `processed/` folder additionally contains `contact_wrench.csv`, `contact_wrench_at_base.csv`, and `wand_pose.csv`. Both the raw and processed datasets are released.

3 Data Records

The dataset is hosted on figshare at <https://doi.org/10.6084/m9.figshare.31990188> [14]. The repository is organized into three top-level subsets: **subset_1** (quasi-static, no contact), **subset_2** (dynamic, no contact) and **subset_3** (dynamic, with external contact). Within each batch, sub-folders are named `<trajectory>_<speed>`, where `<trajectory>` $\in$ {`plane_x`, `plane_y`, `star`, `circle`, `Lissajous`} and `<speed>` $\in$ {`slow` ($\omega_{\max} = 1$ rad/s), `fast` ($\omega_{\max} = 3$ rad/s)}. Every experiment folder contains the raw sensor CSV files, a `processed/` sub-folder with data filtered and uniformly resampled at 100 Hz (time origin $t = 0$ at the first motor command), and a `gvs/` sub-folder with GVS forward-dynamics simulation outputs. All raw CSV files carry a header row with explicit column names and units; timestamps are wall-clock seconds drawn from a single OS clock shared across all sensor processes. Table 4 documents every file type, its columns, units, and content.

Table 4: Complete file and column documentation for each experiment folder. Subscript $n \in \{1, 2, 3, 4\}$ denotes motor index (motors 1 & 3 form the x -plane pair; 2 & 4 the y -plane pair). Subscript $k \in \{0, \dots, 4\}$ denotes disk index from base to tip. Raw files include a header row.

File	Column(s)	Unit	Description
<code>dataMotor.csv</code>	<code>timestamp</code>	s	CPU wall-clock (POSIX)
	<code>phase_index</code>	–	Trajectory phase (0 = idle; successive integers = motion segments)
	<code>homen</code> ($n = 1-4$)	bool	Homing completed flag, motor n

	<code>targetn_rad</code>	rad	Commanded shaft angle, motor n
	<code>abs_anglen_rad</code>	rad	Absolute encoder reading, motor n
	<code>rel_anglen_rad</code>	rad	Relative angle from home, motor n
	<code>velocityn_rad/s</code>	rad/s	Shaft angular velocity, motor n
	<code>torquen_Nm</code>	N m	Output torque, motor n
<code>dataMark10_*.csv</code> (4 files: +x, -x, +y, -y)	<code>timestamp</code>	s	CPU wall-clock (POSIX)
	<code>tension (N)</code>	N	Tendon tension; file suffix indicates tendon direction
<code>dataATIFT.csv</code>	<code>timestamp</code>	s	CPU wall-clock (POSIX)
	<code>Fx (N), Fy (N), Fz (N)</code>	N	Base reaction force along x, y, z
	<code>Tx (Nm), Ty (Nm), Tz (Nm)</code>	N m	Base reaction torque about x, y, z
<code>dataOptiTrack.csv</code>	<code>timestamp_s</code>	s	CPU wall-clock (POSIX)
	<code>diskk_frame_number</code>	–	Motive software frame counter
	<code>diskk_is_valid</code>	bool	Tracking validity (0 = occluded)
	<code>diskk_x/y/z</code>	m	3-D position of disk k
	<code>diskk_q[xyzw]</code>	–	Orientation quaternion, scalar-last convention ($k = 0-4$ for robot disks; in subset 3 also $k = 5$ for the contact wand)
<code>dataFBGS.csv</code>	<code>Timestamp</code>	s	CPU wall-clock (POSIX)
	<code>Curvature_0–_25 (26)</code>	1/m	Curvature at each of the 26 fiber gratings
	<code>Angle_0–_25 (26)</code>	rad	Bending angle at each grating
	<code>Shape_0–_1505 (1 506)</code>	m	Flattened $[x, y, z]$ triplets of the reconstructed 3-D backbone shape at 1 mm arc-length resolution (ILLumiSense Shape CORE output)
<code>dataResenseFT.csv</code> (subset 3 only)	<code>timestamp (s)</code>	s	CPU wall-clock (POSIX)
	<code>Fx, Fy, Fz</code>	N	Contact force along x, y, z
	<code>Tx, Ty, Tz</code>	N m	Contact torque about x, y, z

Table 5: Processed file and column documentation for each experiment folder. Subscript $n \in \{1, 2, 3, 4\}$ denotes motor index (motors 1 & 3 form the x -plane pair; 2 & 4 the y -plane pair). Subscript $k \in \{0, \dots, 4\}$ denotes disk index from base to tip. Processed files are header-less.

File	Column(s)	Unit	Description
<code>angles.csv</code>	col 1: time	s	Time
	cols 2–5: $\theta_1, \theta_2, \theta_3, \theta_4$	rad	Motor shaft angles relative to home
<code>cable_tensions.csv</code>	col 1: time	s	Time
	cols 2–5: $T_{+x}, T_{+y}, T_{-x}, T_{-y}$	N	Low-pass filtered tendon tensions
<code>base_wrench.csv</code>	col 1: time	s	Time
	cols 2–4: F_x, F_y, F_z	N	Base reaction force
	cols 5–7: T_x, T_y, T_z	N m	Base reaction torque;
<code>mocap_frames.csv</code>	col 1: time	s	Time
	x_k, y_k, z_k ($k = 0-4$)	m	Disk position in robot base frame
	$q_{x,k}, q_{y,k}, q_{z,k}, q_{w,k}$	–	Disk orientation quaternion
<code>fbgs_shapes.csv</code>	col 1: time	s	Time
	cols 2–175: $[x_i, y_i, z_i]$, $i = 0-57$	m	Reconstructed 3-D backbone at 1 mm resolution, aligned to robot base frame
<code>contact_wrench.csv</code> (subset 3 only)	col 1: time	s	Time

Continued on next page

Table 5 (continued)

File	Column(s)	Unit	Description
wand_pose.csv (subset 3 only)	cols 2–4: F_x, F_y, F_z	N	Low-pass filtered contact force
	cols 5–7: T_x, T_y, T_z	N m	Low-pass filtered contact torque
	col 1: time	s	Time
	x, y, z	m	Contact wand position in robot base frame
	q_x, q_y, q_z, q_w	–	Contact wand orientation quaternion

3.1 Dataset Structure

The dataset is organized in a hierarchical directory tree. At the top level, four directories separate the experimental conditions and the reference recordings:

```

data/
  quasi_static/
    static_bend_x_100/    ... static_bend_x_200/
    static_bend_x_n100/  ... static_bend_x_n200/
    static_bend_y_100/    ... static_bend_y_200/
    static_bend_y_n100/  ... static_bend_y_n200/
  dynamic_motion/
    plane_x_slow/  plane_x_fast/
    plane_y_slow/  plane_y_fast/
    star_slow/     star_fast/
    circle_slow/   circle_fast/
    Lissajous_slow/ Lissajous_fast/
  contact_motion/
    touch_base/           push_retract/
    plane_y_contact_midway/ plane_y_poke_y/  plane_y_poke_x/
  references/
    released_config/
    straight_config/

```

Each experiment folder (e.g., `circle_slow/`) contains the raw sensor CSV files at the top level and a `processed/` subfolder with the filtered, resampled data:

```

<experiment>/
  dataMotor.csv      dataATIFT.csv
  dataMark10_+x.csv  dataMark10_-x.csv
  dataMark10_+y.csv  dataMark10_-y.csv
  dataFBGS.csv       dataOptiTrack.csv
  processed/
    angles.csv        cable_tensions.csv
    base_wrench.csv   base_wrench_raw.csv
    mocap_frames.csv  fbgs_shapes.csv
    RMSEs.txt         figures/

```

The `quasi_static/` subset contains 16 experiments corresponding to planar bends along $\pm x$ and $\pm y$ at four commanded amplitudes (*i.e.* max rotation angles: 100, 150, 180, 200 °). The `dynamic_motion/` subset contains 10 experiments (5 trajectories $\times$ 2 speeds). The `contact_motion/` subset contains 5 experiments; without actuation `touch_base` and `push_retract`; slow speed only: `plane_y_contact_midway`, `plane_y_poke_y`, and `plane_y_poke_x`. The `references/` folder stores two calibration recordings: `released_config` (tendons released, robot at rest) and `straight_config` (robot held straight with laser alignment).

3.2 Experimental Conditions

Quasi-static subset. The 16 recordings in `quasi_static/` are obtained by driving one pair of antagonist tendons very slowly, so that inertial and damping effects are negligible and the robot reaches a sequence of near-equilibrium configurations. Bending is applied independently along $+x$, $-x$, $+y$, and $-y$, each at four commanded amplitude levels corresponding to maximum tip angles of 100, 150, 180, and 200°. Each recording sweeps the bending amplitude in small increments, producing synchronized measurements of tendon tensions, encoder angles, OptiTrack disk poses, and FBGS curvatures across a range of near-equilibrium configurations.

Dynamic motion subset. The 10 recordings in `dynamic_motion/` cover five trajectory types executed at two speeds ($\omega_{\max} = 1$ rad/s and $\omega_{\max} = 3$ rad/s), producing a range of dynamic regimes.

Planar trajectories (`plane_x`, `plane_y`) drive a single antagonist cable pair, confining the motion to either the xz - or yz -plane. These recordings serve as controlled baselines: the actuation is the simplest possible (one degree of freedom) and the expected robot shape is a planar arc.

Star trajectory (`star`) alternates between the x - and y -cable pairs in successive segments, sweeping the tip through four quadrants in a star-like pattern. This combines both actuator pairs within a single recording, introducing transitions between bending planes and richer coupling between the two actuation channels.

Circular and Lissajous trajectories (`circle`, `Lissajous`) drive both cable pairs simultaneously with sinusoidal commands that are 90 out of phase (`circle`) or with a frequency ratio of 2:1 (`Lissajous`). Both produce continuous out-of-plane and three-dimensional tip motion with sustained coupling between the x and y channels. The Lissajous trajectory additionally changes direction multiple times per cycle, generating more varied accelerations and richer spectral content than the circle. These two trajectories provide the greatest dynamic richness in the dataset.

For all trajectories, the slow speed ($\omega_{\max} = 1$ rad/s) produces moderate dynamic effects, while the fast speed ($\omega_{\max} = 3$ rad/s) amplifies inertial and damping contributions.

Contact motion subset. The five recordings in `contact_motion/` extend the dataset with external contact scenarios. All experiments use an instrumented wand — a rigid rod carrying the Resense contact force sensor at its tip and an OptiTrack marker frame on its body — so that the applied contact force vector and the 6-DoF wand pose are recorded alongside all other robot sensors. Contact experiments are restricted to the planar- y trajectory because the geometrical constraints of the experimental setup — the thin spacer disks, the OptiTrack marker brackets attached to them, and the small tip area of the Resense sensor — make reliable and repeatable contact difficult to achieve on more complex three-dimensional motions.

`touch_base` is a sensor cross-validation recording with no actuation and no robot motion. The wand is pressed against the robot base while the ATI-FT sensor simultaneously measures the reaction wrench at the base. This controlled static loading provides a direct comparison between the Resense contact force and the ATI-FT base force, serving as an independent cross-check of both sensors (see Section 4).

`push_retract` records the response of the unactuated robot to an external tip load. The tendons are fully released so the robot rests in its natural unloaded configuration, and the wand is slowly pushed against the tip then retracted. The simultaneous recording of the Resense contact force and the ATI-FT base wrench during this controlled loading provides a second cross-validation point for the two force sensors, complementing `touch_base` where the load was applied at the base rather than the tip.

`plane_y_contact_midway` is an actuated experiment in which the robot executes the planar- y trajectory (slow speed) while the wand is held fixed against the spacer disk located 18 cm from the base, blocking the robot’s motion at that point. The robot continues to receive motor commands against the obstacle, so tendon tensions, base wrench, and FBGS curvatures all reflect the redistribution of internal forces caused by a mid-body obstruction. This scenario is representative of contact during confined-space manipulation.

`plane_y_poke_y` is a dynamic perturbation experiment in which the robot moves on the planar- y trajectory (slow speed) and the wand delivers brief transverse pokes in the y -direction — the primary bending direction of the trajectory. Each poke superimposes a transient oscillation on the nominal motion.

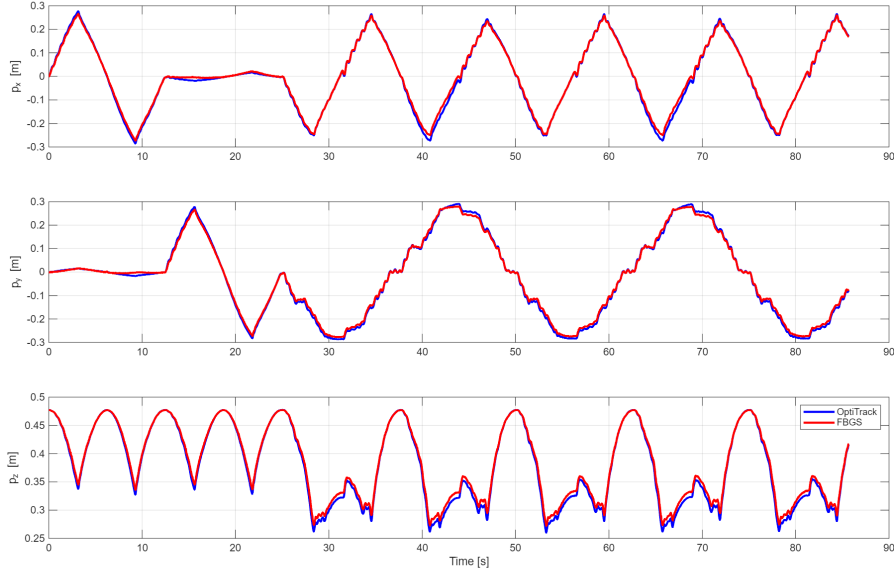


Figure 4: Trajectory of the robot tip as measured by the OptiTrack motion capture system (blue) and the FBGS (red) for the Lissajous trajectory.

`plane_y_poke_x` mirrors the previous experiment with pokes delivered in the x -direction, orthogonal to the nominal bending plane. These pokes excite out-of-plane oscillations and couple the x and y dynamics. Together, `plane_y_poke_y` and `plane_y_poke_x` cover both in-plane and out-of-plane perturbation directions.

3.3 Intended use per subset

Each subset is designed to support a distinct class of research tasks.

The quasi-static subset (`subset_1`) provides steady-state configurations at known tendon tensions and motor displacements. It is suited for mechanical parameter identification: the bending rigidity EI of the backbone can be estimated by fitting a static rod model to the OptiTrack disk poses under the measured Mark-10 tensions, as described in [Section 4.4.1](#). In addition, the simultaneous recording of motor shaft angles and cable tensions at multiple load levels enables direct estimation of cable axial stiffness from the force–displacement relationship $k_c = \Delta T / \Delta \ell$, where $\Delta \ell = r_s \Delta \theta$ is derived from the encoder readings.

The dynamic subset (`subset_2`) covers a range of trajectories and speeds with full multi-sensor recording. It is suited for state estimation (fusing OptiTrack, FBGS, and encoder measurements), system identification of dynamic parameters (distributed mass, damping), and benchmarking of data-driven or model-based control and prediction methods.

The contact subset (`subset_3`) extends the dynamic recordings with contact force measurements from the Resense sensor and the 6-DoF pose of the contact object from OptiTrack. It is suited for contact dynamics modeling, validation of wrench transmission models along the robot body, and benchmarking of contact detection and force estimation algorithms.

4 Technical Validation

This section presents the analyses performed to verify the technical quality and internal consistency of the recorded dataset. The dataset contains two families of measurements: kinematic quantities (OptiTrack poses, FBGS strain and reconstructed shape, motor encoder angles) and force quantities (Mark10 cable tensions, ATI-FT base wrench, Resense contact force). Within each family, sensors may be directly compared. Kinematic sensors are cross-validated by comparing OptiTrack tip positions with FBGS-reconstructed shapes, and encoder-derived cable displacements with motion-capture-based geometric

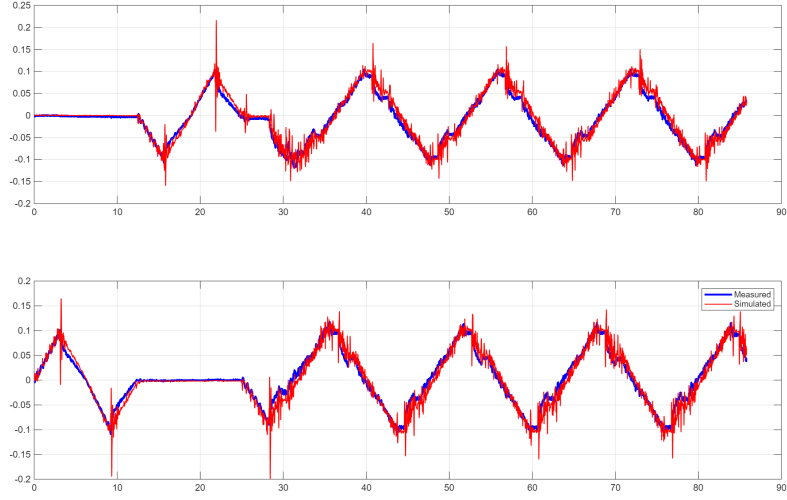


Figure 5: Comparison of the simulated torque at the base (red) with the measured torque by the (unfiltered) ATI FT sensor (blue) using the circular motion.

reconstructions. On the other hand, force measurements cannot be compared directly without a physical model; therefore, a reduced-order dynamics model is employed as a cross-validation tool to verify force-kinematics consistency, as detailed in Section 4.4.

4.1 Sensor noise and calibration

A dedicated static recording (`references/released_config`) was collected with the robot hanging at rest and all tendons released. This recording provides a sensor-independent characterisation of the noise floor, decoupled from any robot motion. In addition, the robot remains static for the first ~ 3 s of every experiment before the motors begin to move; this preamble can be used to compute a per-experiment sensors bias.

Their output is quantized at the sensor resolution of 0.02 N; no variation was observed in the static recording, confirming that the noise floor is below the resolution.

Table 6 reports the standard deviation of each sensor measured over the full reference recording.

4.2 Kinematic cross-validation

4.2.1 Comparison of recorded shape with FBGS and Motion Capture

The OptiTrack and FBGS sensors provide independent measurements of the robot shape. To verify their mutual consistency, we compare the tip position recorded by OptiTrack with the tip position reconstructed from FBGS strain data, evaluated at the same arc-length coordinate. Figure 4 shows the overlay of these two signals for the `Lissajous` trajectory (see Fig. 3e). The overlay shows good consistency between the two measurements; residual deviations are attributable to the frame definition uncertainties discussed in Section 4.1. Table 7 reports the root-mean-square error (RMSE) between the OptiTrack and FBGS tip positions for all dynamic trajectories. The tip RMSE remains below 12 mm in the primary motion axes and below 8 mm in the vertical (z) direction across all trajectories and speeds, confirming that the two independent kinematic measurements are mutually consistent.

4.2.2 Validation of Cable Displacement via Motion Capture

An independent geometric estimate of the cable displacement is derived from the motion-capture data to cross-validate the encoder-based measurements recorded by the actuators.

To obtain a more accurate description of the cable path, the five measured disk frames are interpolated to obtain the poses of the intermediate disks (not provided with mocap frames). Positions are interpolated

Table 6: Static noise characterisation. Values are standard deviations computed over the reference recording with the robot at rest. Mark-10 gauges output a quantised reading at 0.02 N resolution; no variation was observed in the static recording.

Sensor	Quantity	Static noise (1σ)
Mark-10 (+x)	Cable tension	< 0.02 N
Mark-10 (+y)	Cable tension	< 0.02 N
Mark-10 (-x)	Cable tension	< 0.02 N
Mark-10 (-y)	Cable tension	< 0.02 N
ATI-FT	F_x	0.013 N
	F_y	0.017 N
	F_z	0.944 N
	T_x	7.1×10^{-4} N m
	T_y	4.3×10^{-4} N m
	T_z	7.9×10^{-3} N m
OptiTrack disk 0	Position (x, y, z) [mm]	(0.005, 0.005, 0.003)
	Orientation (X, Y, Z) [deg]	(0.0043, 0.0034, 0.0049)
OptiTrack disk 1	Position (x, y, z) [mm]	(0.005, 0.005, 0.004)
	Orientation (X, Y, Z) [deg]	(0.0022, 0.0029, 0.0032)
OptiTrack disk 2	Position (x, y, z) [mm]	(0.007, 0.008, 0.008)
	Orientation (X, Y, Z) [deg]	(0.0049, 0.0126, 0.0071)
OptiTrack disk 3	Position (x, y, z) [mm]	(0.008, 0.008, 0.003)
	Orientation (X, Y, Z) [deg]	(0.0034, 0.0032, 0.0049)
OptiTrack disk 4	Position (x, y, z) [mm]	(0.011, 0.013, 0.005)
	Orientation (X, Y, Z) [deg]	(0.0068, 0.0109, 0.0077)
FBGS	Avg. shape noise along arc (x, y, z) [mm]	(0.002, 0.194, 0.233)
	Tip position noise (x, y, z) [mm]	(0.006, 0.476, 0.508)

with a shape-preserving piecewise-cubic Hermite (PCHIP) scheme, while orientations are interpolated by converting each rotation matrix to a unit quaternion, applying PCHIP independently to each quaternion component, and normalizing the result to the unit sphere. The consistency of the quaternion hemisphere is enforced between successive disks to avoid sign-flip artifacts. At each interpolated frame the absolute position of every cable is computed by applying the local offset vector $\mathbf{d}_c$ through the interpolated rotation matrix. The total length of cable ℓ_c is then approximated as the piecewise-linear sum of Euclidean distances between consecutive attachment points along the interpolated backbone.

The cable displacement is defined as the change in **length relative to the straight (pre-motion) configuration**, estimated by averaging the first $N_{\text{ref}} = 10$ frames:

$$\Delta \ell_c(t) = \ell_c(t) - \bar{\ell}_c^{\text{ref}}, \quad \bar{\ell}_c^{\text{ref}} = \frac{1}{N_{\text{ref}}} \sum_{k=1}^{N_{\text{ref}}} \ell_c(t_k). \quad (1)$$

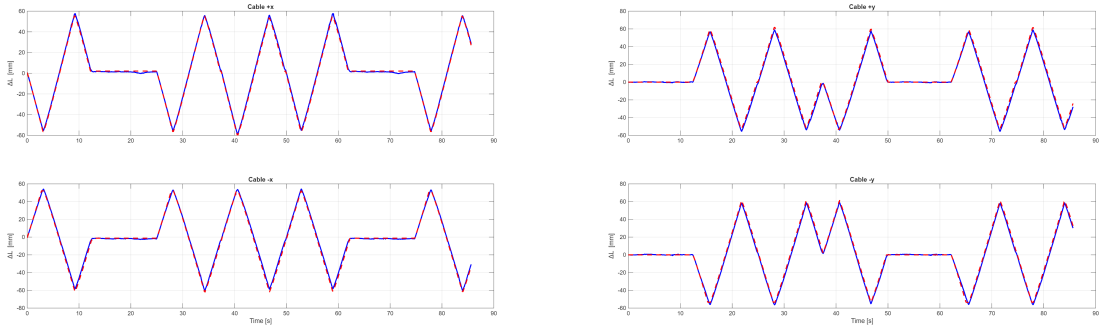


Figure 6: Cable displacement as measured by the actuator encoders (red) and reconstructed from the OptiTrack kinematics (blue) for the star trajectory: x pair (left) and y pair (right).

Table 7: RMSE of tip position between OptiTrack and FBGS for each dynamic trajectory. Values are in millimeters; percentages in parentheses are relative to the motion range.

Trajectory	RMSE x [mm] (%)	RMSE y [mm] (%)	RMSE z [mm] (%)
plane_x slow	8.01 (1.4%)	N/A	3.62 (2.5%)
plane_x fast	9.88 (1.7%)	N/A	5.28 (3.4%)
plane_y slow	N/A	9.04 (1.6%)	3.99 (2.7%)
plane_y fast	N/A	10.27 (1.8%)	5.48 (3.5%)
star slow	7.36 (1.3%)	7.86 (1.4%)	5.34 (1.9%)
star fast	8.95 (1.5%)	8.75 (1.5%)	6.85 (2.3%)
circle slow	11.24 (1.9%)	8.98 (1.6%)	6.94 (4.0%)
circle fast	11.48 (2.0%)	9.67 (1.6%)	7.41 (4.0%)
Lissajous slow	9.21 (1.6%)	8.51 (1.5%)	6.50 (3.0%)
Lissajous fast	10.59 (1.8%)	9.07 (1.6%)	7.26 (3.1%)

On the other hand, an independent estimate is obtained from the motor encoders as $\Delta\ell_c^{\text{enc}}(t) = (\theta_c(t) - \theta_c^{\text{ref}}) r_s$, where θ_c is the measured shaft angle and r_s is the spool radius (see Table 2). The two estimates are compared directly in the time domain; agreement between them constitutes a geometric consistency check, confirming that the encoder records and the motion-capture poses are mutually consistent and that the assumed cable-routing geometry is correct. Figure 6 compares the two estimates for the star trajectory. Table 8 reports the RMSE between encoder-based and motion-capture-based cable displacement for all dynamic trajectories. The cable RMSE remains below 4.2 mm across all trajectories, corresponding to less than 3.6% of the motion range, confirming that the encoder recordings and the motion-capture poses are geometrically consistent.

Table 8: RMSE of cable displacement between encoder and OptiTrack reconstruction for each dynamic trajectory. Values are in millimeters; percentages in parentheses are relative to the cable motion range. Cables are labeled by tendon direction ($+x$, $+y$, $-x$, $-y$).

Trajectory	$+x$ [mm] (%)	$+y$ [mm] (%)	$-x$ [mm] (%)	$-y$ [mm] (%)
plane_x slow	1.85 (1.7%)	N/A	3.57 (3.1%)	N/A
plane_x fast	2.43 (2.2%)	N/A	3.53 (3.1%)	N/A
plane_y slow	N/A	2.87 (2.5%)	N/A	2.44 (2.1%)
plane_y fast	N/A	2.77 (2.5%)	N/A	3.05 (2.6%)
star slow	1.41 (1.2%)	2.31 (2.0%)	1.93 (1.7%)	2.09 (1.8%)
star fast	1.88 (1.6%)	2.78 (2.5%)	2.20 (1.9%)	2.29 (2.0%)
circle slow	2.67 (2.3%)	3.74 (3.2%)	3.00 (2.6%)	3.12 (2.7%)
circle fast	3.15 (2.7%)	4.17 (3.6%)	3.32 (2.9%)	3.29 (2.8%)
Lissajous slow	1.22 (1.1%)	2.38 (2.0%)	1.73 (1.5%)	2.15 (1.8%)
Lissajous fast	1.93 (1.7%)	2.58 (2.2%)	2.02 (1.7%)	2.18 (1.9%)

4.3 Validation of contact wand pose

The contact wand is provided with a mocap frame to track its movement in the workspace. To ensure the correct position of the sensor is recorded, we compared its position with the one of the robot tip (last disk). Figure 7 shows the position of the robot tip and the sensor at the contact wand. We know that the Euclidean distance between the sensor and the robot tip is ≈ 38 mm which is consistent with the dimension of the disks.

4.4 Force-kinematics consistency

Cable tensions (measured with Mark10 sensors) and base wrench (measured with ATI-FT sensor) cannot be directly compared without a physical model relating forces to deformations. To verify that these two families of measurements are mutually consistent, we employ a forward dynamics simulation based on the Geometric Variable Strain (GVS) approach [5] as a cross-validation tool.

As the GVS model requires the parameters of the robot to be known, the next section will address how

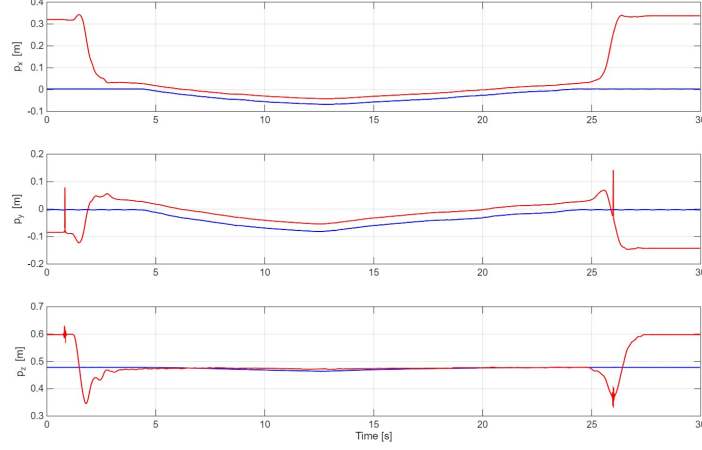


Figure 7: Position xyz of the robot tip (blue) and the Resense sensor attached at the contact wand (red) during the push/retract motion.

these parameters were roughly estimated. Then, the following section discusses the comparison of the simulation results with the recorded data.

4.4.1 Rod mechanical parameter characterization

An equivalent bending rigidity EI of the backbone was estimated with the quasi-static subset, where the robot was commanded to planar bending in the xz and yz planes.

To avoid short settling effects after actuation, the first ~ 200 samples of each recording were removed and the remaining samples were averaged, yielding disk positions $\{\mathbf{p}_i^{\text{meas}}(w)\}_{i=1}^{N_{\text{disks}}}$ in the base frame. For a candidate value of EI , a static Cosserat-rod solver was used to predict the equilibrium shape under the measured tensions $\boldsymbol{\tau}_w$ by enforcing the boundary conditions through a root-finding routine. The simulated disk positions evaluated at the same arc-length locations are denoted by $\mathbf{p}_i^{\text{sim}}(EI, \boldsymbol{\tau}_w)$.

For each bending mode (plane), EI was selected by minimizing the aggregated geometric discrepancy between simulated and measured disk positions over the corresponding set of trials $\mathcal{W}$:

$$\hat{EI} = \arg \min_{EI} \sum_{w \in \mathcal{W}} \sum_{i=1}^{N_{\text{disks}}} \|\mathbf{p}_i^{\text{sim}}(EI, \boldsymbol{\tau}_w) - \mathbf{p}_i^{\text{meas}}(w)\|_2^2. \quad (2)$$

The resulting plane-wise estimates were $\hat{EI}_{xz} = 0.0925$ and $\hat{EI}_{yz} = 0.08875$, a single value $\hat{EI} = 0.092 \text{ N} \cdot \text{m}^2$ was used in the remainder of this work.

Then, the distributed weight of the rod is calculated from the measured weight of the robot (0.102 Kg) and the geometry of the rod (length 0.48 m and radius 2 mm) producing $\rho = 16910 \text{ kg/m}^3$.

Table 9: Mechanical parameters of the continuum robot, * means the quantity has been computed via optimization.

Stiffness* $\hat{EI}$	0.092 N m ²
Robot mass	0.102 kg
Backbone length	0.48 m
Equivalent density ρ	16 910 kg m ⁻³

4.4.2 Comparison of the measured forces

Using the GVS of [5] and the characterized robot materials we have a reliable framework to compute the consistency of the measured values. Using the tendon tension as input for our simulation, we can

Table 10: RMSE between GVS-simulated and ATI-FT-measured base torque for each dynamic trajectory. Values are absolute (N m) with the percentage of the signal range in parentheses. The two components T_x and T_y correspond to the bending moments about the x and y axes at the robot base.

Trajectory	Speed	RMSE T_x [N m] (%)	RMSE T_y [N m] (%)
plane_x	slow	N/A	0.0162 (7.8%)
	fast	N/A	0.0474 (19.4%)
plane_y	slow	0.0160 (7.6%)	N/A
	fast	0.0417 (17.0%)	N/A
star	slow	0.0141 (6.4%)	0.0138 (6.2%)
	fast	0.0342 (13.9%)	0.0362 (15.2%)
circle	slow	0.0155 (6.9%)	0.0163 (7.2%)
	fast	0.0357 (13.7%)	0.0391 (15.2%)
Lissajous	slow	0.0148 (6.3%)	0.0147 (7.0%)
	fast	0.0389 (17.6%)	0.0387 (15.8%)

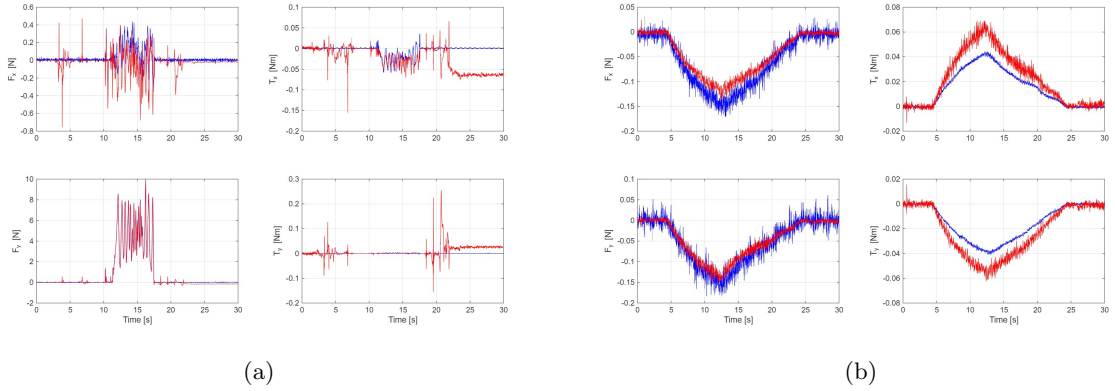


Figure 8: Comparison of the measured forces (F_x and F_y) and torques (T_x and T_y) from the measurement of the Resense to the ATI FT sensors for the touch_base (a) and push_retract (b) recordings.

compute the corresponding motion of the robot and the forces it exchanges at its base. Figure 5 compares the simulated and measured base torque. The overall trend is well captured; residual discrepancies (*e.g.*, torque spikes) are attributed to approximate identification of the distributed mass and inertia parameters (Section 4.4.1). Table 10 reports the RMSE between the GVS-simulated and ATI-FT-measured base torque components T_x and T_y across all dynamic trajectories. For slow trajectories, the RMSE remains below 0.02 N m, corresponding to less than 8% of the signal range for the primary bending axis of each motion. At fast speed, the error increases to ~ 0.04 N m (~ 15 – 20%), consistent with the approximate inertial parameters used in the simulation (Section 4.4.1). Overall, the qualitative agreement between simulation and measurement across all trajectories confirms the internal consistency of the force and kinematic recordings.

4.5 Contact force validation

The Resense HEX 12 contact sensor was validated through two static tests that verify the consistency of the measured contact wrench with the ATI-FT base wrench via rigid-body wrench transformation. In both tests, all tendons were released so that the only external load on the robot is the contact force applied by the wand (plus gravity).

In the first test, the contact wand was pressed against the robot base disk along the y -axis of the base motion-capture frame, so that the lever arm between the contact point and the ATI-FT sensing origin is small and well-defined. The wrench measured by the Resense sensor was mapped into the ATI-FT coordinate frame using the adjoint map $\text{Ad}_{g_{ba}}$, computed from the OptiTrack-measured relative pose $g_{ba} \in \text{SE}(3)$ between the base disk and the contact wand:

$$\mathcal{W}_{\text{contact}}^{\text{base}} = -\text{Ad}_{g_{ba}} \mathcal{W}_{\text{Resense}}^{\text{wand}}, \quad (3)$$

where the minus sign accounts for the action-reaction principle. The adjoint map is a purely geometric operation that transports a wrench between two coordinate frames; it does not assume rigidity of the body connecting the frames, and therefore remains valid even when the continuum robot deforms between the contact point and the base. Figure 8a compares the transformed Resense wrench with the ATI-FT measurement. The two measurements show close agreement, confirming that the Resense calibration and the geometric transformation are mutually consistent.

In the second test, the wand was used to push the robot at its tip while the robot undergoes a quasi-static deformation. The Resense wrench was again mapped into the base frame using Eq. (3), now with the full pose chain from the wand to the base (a lever arm of up to 0.48 m). Figure 8b compares the transformed forces and torques with the corresponding ATI-FT readings. The force components show good agreement, while the torque components exhibit small deviations that grow with the lever arm, consistent with the compounding effect of gravity that is applied on the Resense sensor.

4.6 Temporal synchronization

All sensors are acquired as concurrent threads in a single Python process on a single machine. Each thread timestamps its readings using the operating system wall-clock (`time.time()` in Python), ensuring that all sensors share the same clock source without requiring hardware synchronization. The resulting synchronization error is bounded by the operating system thread scheduling jitter, which is on the order of 1 ms to 2 ms on the acquisition machine, well below the signal periods of interest (the highest dynamic content in the dataset is below 20 Hz).

5 Usage Notes

Software requirements. The raw and processed data are stored as plain CSV files and can be loaded in any programming language. The post-processing pipeline (`process_data.m`) was developed and tested in MATLAB R2025b, using only the built-in functions `butter`, `filtfilt`, and `interp1` from the Signal Processing Toolbox. Users who prefer Python can replicate the processing with `scipy.signal.butter`, `scipy.signal.filtfilt`, and `scipy.interpolate.interp1d`, or use `pandas.read_csv` to load the data directly.

Known limitations and caveats.

- **FBGS coordinate alignment.** The FBGS fiber cannot be bonded to the backbone with sub-degree angular accuracy. A residual rotational offset between the fiber deformation plane and the robot bending plane is corrected in post-processing using the initial planar calibration sweeps (see Methods). Users working with the raw FBGS data must apply this correction themselves.
- **ATI-FT F_z noise.** The ATI Mini40 exhibits a high noise floor on the F_z axis ($1\sigma \approx 0.94$ N), which is an intrinsic characteristic of this sensor and calibration range. Users should account for this when using the base wrench data.
- **OptiTrack marker uncertainties.** Definition of frames with the OptiTrack markers suffer from uncertainties in the markers placement and assembly.
- **Motor trajectory generation.** The motors are position-controlled; the trajectories are defined as sequences of waypoint angles with a maximum angular velocity limit. The actual tip trajectory depends on the robot dynamics and is not a perfect geometric curve.

6 Data Availability

The dataset described in this work is publicly available on figshare at <https://doi.org/10.6084/m9.figshare.31990188> [14]. The repository contains all raw and processed sensor recordings organized as described in the Data Records section (Section 3).

7 Author Contributions

8 Competing Interests

The authors declare no competing interests.

9 Funding

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